

CE 310 — Week 11 Weekly Quiz

Week: 11 | **Topic:** Hypothesis Testing — t-Tests and ANOVA

Due: Closes **8:00 AM Tuesday, Nov 10** — the day before Exercise 11 and Homework 11 are due. Taken on D2L

Points: 15 questions × 1 pt each = 15 pts

Attempts: Two — the higher score counts

Time limit: 25 minutes

Instructions: Answer each question in the format it asks for — most are multiple choice, with one matching, one ordering, one select-all-that-apply and one true/false. This quiz is auto-graded and must be submitted before it closes at the time given above. You may use your notes and readings, but work independently.

This quiz covers the Monday lecture and the Wednesday Python tutorial — null and alternative hypotheses, p-values, Type I and II errors, the t-test, ANOVA, effect size, and the Python functions that go with them. The tutorial's warning boxes are where the most time-consuming mistakes are flagged, so they are worth reading before you start.

Questions

Q1. [Matching] Match each term to what it means in a hypothesis test.

Choices:

- (A) H_0 , the null hypothesis
- (B) α , the significance level
- (C) The p-value
- (D) The power of the test
- (E) A Type II error

Matches:

1. How surprising the observed data would be if H_0 were true
 2. The statement of no effect that the test sets out to disprove
 3. Failing to reject H_0 when it is in fact false
 4. The false-positive rate you agree to accept, fixed before the test is run
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Q2. [True/False] A p-value of 0.03 means there is a 3% probability that the null hypothesis is true.

Q3. A two-sample t-test compares 555 winning bids against 3,814 losing bids on the same line item. How many degrees of freedom does the test have?

- A) $555 - 1$, because the winning group is the one under test
 - B) 4,369 — one for every bid in the dataset
 - C) $555 + 3,814 - 2$, because a mean was estimated for each group
 - D) It cannot be determined without knowing each group's variance
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Q4. [Select all that apply] Which of the following are true of statistical significance?

- (A) With a large enough sample, an effect too small to matter can still come out statistically significant
 - (B) A significant result establishes that the effect is large
 - (C) An effect size such as Cohen's d speaks to practical importance in a way a p-value does not
 - (D) Failing to reject H_0 establishes that H_0 is true
 - (E) α should be chosen before the p-value is known
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Q5. The t-statistic in a two-sample t-test is approximately:

- A) The difference in group means divided by the standard error of that difference
 - B) The product of each group's sample size
 - C) The ratio of the sample variance to the population variance
 - D) The square root of the F-statistic from a one-way ANOVA
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Q6. A one-way ANOVA is the appropriate test when you want to:

- A) Determine whether a single sample mean equals a hypothesised value
 - B) Compare means from three or more groups while controlling the Type I error rate
 - C) Compare exactly two group means when the two samples are independent of each other
 - D) Test whether two variables are linearly correlated
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Q7. [Ordering] Put the steps of a two-sample hypothesis test in the order you would carry them out.

- (A) State H_0 and H_1 for the comparison you actually care about
- (B) Choose α before looking at the data
- (C) Compute the test statistic and its p-value
- (D) Compare p against α , and report an effect size alongside the verdict

Q8. Cohen's d is a measure of:

- A) The probability of rejecting H_0 when H_1 is true (statistical power)
- B) The ratio of the F-statistic to the degrees of freedom in an ANOVA
- C) The fraction of variance explained by a regression model
- D) The standardised difference between two group means, in standard-deviation units

Q9. A dataset has $n = 8,760$ observations. You compute a t-statistic of 2.1 with $p = 0.036$, and Cohen's $d = 0.05$. Which statement best describes this result?

- A) The effect is statistically significant *and* practically large
- B) The effect is not statistically significant because Cohen's d falls below the conventional 0.2 threshold for a small effect
- C) The effect is statistically significant but practically very small ($d < 0.2$ is conventionally "negligible")
- D) The result is uninterpretable without knowing the population standard deviation

Q10. In Python, `scipy.stats.f_oneway(group1, group2, group3)` does which of the following?

- A) Fits a linear regression with three predictor variables
- B) Computes three separate t-tests, one for each pair of groups
- C) Returns the variance of each group separately
- D) Runs a one-way ANOVA and returns the F-statistic and p-value

Q11. `t_stat, p_val = stats.ttest_ind(winners, losers)`. What is `p_val`?

- A) The probability that the null hypothesis is true, given the observed data
- B) The probability of a difference this large if the means were truly equal
- C) The probability that the two population means are genuinely different
- D) The size of the difference between the two means, in the original units

Q12. A t-test on 8,760 hours returns $p < 0.001$, and Cohen's d is 0.08. What do you conclude?

- A) The difference is real but tiny; a big sample makes almost anything significant

- B) A very strong effect, since p is so far below the conventional 0.05 threshold
 - C) The test is invalid, because Cohen's d has come out below the 0.2 floor
 - D) The two results contradict each other, so the whole test should be re-run
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Q13. An ANOVA across four line items returns $p = 0.002$, and you then call `pairwise_tukeyhsd`. Why is that second step needed?

- A) To confirm the ANOVA result independently, using a second and different method
 - B) ANOVA says only that some pair differs; Tukey identifies which pair
 - C) To check that the four line items have roughly equal variances before trusting p
 - D) Because ANOVA cannot handle more than three groups at a time without adjustment
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Q14. You run `stats.ttest_1samp(ratio, 1.0)`. What question is that asking?

- A) Whether the mean bid-to-estimate ratio differs from the fixed value 1.0
 - B) Whether the winning-bid mean differs from the losing-bid mean on this item
 - C) Whether any single bid with a ratio of 1.0 is unusual within the sample
 - D) Whether the ratios are Normally distributed around a centre of 1.0
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Q15. A regression summary reports a t -value on each coefficient. What is each one testing?

- A) Whether the model as a whole is significant across all of its predictors
- B) Whether that coefficient differs from zero, given the other predictors
- C) Whether the residuals of the fitted model are Normally distributed
- D) Whether that predictor is correlated with the response on its own